UNIFIED MULTI-TASK DEEP LEARNING FRAMEWORK FOR PRECISION CATTLE MONITORING

BY

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1. ABSTRACT

Weed detection in precision agriculture is hindered due to need of tedious pixel-level annotation and domain shift between test and training data. A number of Visual Foundation Models (VFMs) such as the Segment Anything Model (SAM) have been used to solve the above problems. But its implementation often has a significant computational cost and is prone to semantic confusion between similar classes. In this thesis, a novel label-efficient framework for weed detection is proposed. It combines a lightweight few-shot object detector that will generate spatial prompts for VFMs. This will enable them to achieve high-fidelity zero-shot instance segmentation. Because it relies on an efficient zero-shot detector, it can be deployed quickly in the field without the need of exhaustive crop-specific retraining. The proposed architecture will be a computationally viable universal perception system for scalable and autonomous precision weed management.

2. INTRODUCTION

The global yield loss caused by weeds amounts to 34% on average (Oerke et al). The traditional homogeneous chemical weedicide application and labor-intensive hand weeding is becoming ineffective due to ecological impact, rising costs and the rapid evolution of herbicide-resistant weeds [5]. There is a growing need for sustainable production methods. One such promising method is the development of automated precision agriculture solutions that use machine vision for site-specific precision weed management.

While appealing in theory, deep learning-based approaches are currently limited by their reliance on fully supervised frameworks. These approaches require extensive pixel level annotated dataset, which is a time and resource-consuming process (up to 2.5 hours per image) [9]. Furthermore, the traditional computer vision models suffer from domain shift problem and fail catastrophically when deployed in a new field due to the significant variations in lighting, soil type and crop geometries [26]. In this work, we propose a novel universal zero-shot weed detection framework hybrid foundation-model architecture, which requires only a small amount of annotated data.

2.1 Background of the Study

Recent advancements in agricultural computer vision reveal that one-stage detectors, like YOLO variants, effectively localize multiple targets at a fast speed in natural environments [8]. At the same time, Vision Foundation Models such as the Segment Anything Model (SAM) enable powerful zero-shot segmentation tasks through the use of a pre-trained architecture that does not require task-specific training data [4]. Recent works highlight the possibility of combining such models in a cascade, where a fast, lightweight one-stage detector is used to provide spatial bounding box prompts to a foundation model, achieving superior performance over standard one-stage or two-stage detectors. However, such approaches are often memory-intensive and suffer from a computational complexity of quadratic complexity in the number of prompts, necessitating optimization in order to be deployed on agricultural hardware at the edge [23].

2.2 Scope of the Work

This research will be limited to the development of a software which will be used to use the vision-based system to detect weeds on a robotic rover. The proposed approach will be based on the use of publicly available image sets of different crops (deng et al). The research effort will not include the development of the mechanical design of the rover, nor the development of physical mechanisms that will perform the weeding function.

2.2.1 Assumptions

The current study makes two critical assumptions in order to conduct the analysis. First, the data sets and the camera equipment used for this research will produce standard daylight optical images. Their quality and resolution are sufficient to allow for the generation of spatial prompts accurate enough for the foundational model [4]. Second, the limitations inherent in the edge-computing hardware can be approximated in software via standard performance metrics, such as FPS and parameter count.

3. LITERATURE REVIEW

The development of robust, automated weed detection systems has been a central focus of precision agriculture research over the past decade. Making a universal weed detection framework is hindered by technical challenges, such as environmental variability, lack of annotated data, and the visual similarities between crops and weeds. This section critically examines the use of different computer vision techniques applied to this domain over the years. It progressed from early supervised machine learning approaches to the current state-of-the-art in artificial intelligence.

3.1 Traditional Deep Learning in Precision Agriculture

The shift from traditional machine vision to deep neural networks is one of the recent advances that increased the accuracy of automated weed recognition under controlled environments. Supervised Convolutional Neural Networks and one-stage object detectors, such as Yolo family networks, allow to automatically learn higher-level features from raw pixel data [1] [17]. Recent developments in neural network design have shown outstanding performance in terms of speed and accuracy. According to [2], Yolo v11 achieved 13.5 ms inference speed and 89.10% mAP on average. In turn, Yolov4-Tiny achieved 42.24 FPS on UAV maize weed detection [32]. The accuracy of a custom-made 5-layered Convolutional Neural Network reached 95.12% with only 16.754 ms of inference speed and 0.012 GB memory consumption on a Raspberry Pi platform [3].

However, the accuracy of such methods is limited by the quality and quantity of annotated data, which is a known limitation of traditional computer vision (Deng et al., 2024). To train such a model, one needs to create pixel-level annotations, which is a very time-consuming process. According to (Gao et al., 2025), annotating one image could take 15-20 minutes. Therefore, it is not feasible to collect annotated data for every type of weed on the planet.

In addition to the lack of annotated data, deploying such a model in real-world conditions faces the problem of domain shift. In other words, if the distribution of data where the model is deployed differs from the distribution of data on which it was trained, its accuracy drops significantly (gao et al). The magnitude of this drop was recently quantified in an attempt to classify crops using a Source-Only approach: “On the cross-continental crop classification task, the mAP of a Source-Only model decreased by as much as 33.67% from a value of 99.17% (upper bound) to 65.50% on the target set, completely failing to recognize the “Cotton” class” (Li et al., 2025). Similar results were presented when analyzing seasonal changes: “The baseline YOLOX, YOLOv8 and

DINO detectors experienced performance drops ranging from 8.8%, to 14.5% on the cross-year experiment, with accuracy decreases of up to 26.1% for DINO on the maize classification task" (Deng et al., 2024). Domain shift poses an even bigger challenge when it comes to recognizing plant species in the field. On images with a high degree of overlap and varying lighting conditions, "the mAP of Faster R-CNN fell to as low as 9.55%" (Gao et al., 2025).

3.2 Unsupervised Domain Adaptation (UDA) and Semi-Supervised Learning

In order to save the astronomical amount of manual labor required to annotate a large number of images, and to reduce the catastrophic failures that occur due to domain shift, the agricultural research community has increasingly shifted to Unsupervised Domain Adaptation (UDA) and Semi-Supervised Learning (SSL) [9]. Initial methods based on Generative Adversarial Networks (GANs) and CycleGAN for style transfer produced results that were adequate for getting the source images translated to fit into target domains, with an Intersection over Union (IoU) increase of around 10% on various source-target field pairs for the mean Intersection over Union (mIoU) metric [10] [26]. To fill the significant visual gap between various plant images from the internet and the limited target domain agricultural robot data, Multi-level Attention-based Adversarial Discriminators (MAAD) have been incorporated into feature extractors, which resulted in 7.5% higher object detection precision on unlabeled target domains [25].

Likewise, semi-supervised teacher-student architectures have shown that using large amounts of unlabeled data can greatly improve accuracy. YOLOv8l-based WeedTeacher achieved a 2.6% improvement over the fully supervised baseline with only 10% of the labeled vegetable field (deng et al). However, other semi-supervised methods based on FCOS and Faster R-CNN on multi-class cotton datasets had about 76% and 96% detection accuracy, and greatly reduced the dependency on annotations (li et al). To avoid the model being overfitted by predicting false targets, class covariance analysis based pseudo labelling has been proposed [31]. New feature-level UDA improvements have enabled the development of masked adversarial alignment (CCUDA-SS) to selectively adapt models to uncertain areas while incurring minimal runtime complexity, resulting in up to +5.23% mIoU improvement on eight datasets [9]. Moreover, 80.53% mIoU was achieved by the 1-shot Supervised Domain Adaptation (SDA) model, which was very close to the 83.69% accuracy of the fully supervised network when UDA was combined with the use of augmentation scheduling [10].

But there are inherent limitations of UDA and SSL approaches, which limit their universal application. The most important weakness is "pseudolabel drift," which occurs when the domain changes significantly, and when a label is misclassified at the beginning, it is continually reinforced, actively contaminating the teacher model's target distribution [9]. Empirical results demonstrate that sub-optimal SSOD framework designs can negatively impact performance when domain shift is strong: for example, the DenseTeacher framework was applied to cross-domain weed datasets, yielding a significant decrease in mAP@50 of 13.4% from DenseTeacher's fully supervised baseline (deng et al).

3.3 The Paradigm Shift to Foundation Models and Zero-Shot Learning

Looking beyond the iterative domain adaptation paradigm, the agricultural computer vision arena is witnessing a paradigm shift toward Visual Foundation Models (VFMs). After their introduction to the computer vision domain, vision transformers (ViTs) showed to have clear advantages over local CNNs in terms of capturing global relations [21]. For instance, in a comparison between hierarchical ViTs (e.g. Swin Transformer) and standard ViT-B, the former achieved 92.3% average accuracy on multi-crop settings [12]. As for novel architectures, the proposed WeedSwin model demonstrated outstanding performance for multi-stage classification tasks, attaining a test mAP of 0.993 at 218.27 FPS [13]. Meanwhile, to address the issue of large parameter budgets in most Transformers, a lightweight alternative was suggested, which slashed the number of parameters to 412K while managing to keep a reasonable inference time of 141ms for practical edge-level deployment [19].

Beyond the architecture design space, there has been an increasing number of works leveraging the powerful but simplistic Segment Anything Model (SAM) from Meta. For instance, the zero-shot adaptation ability of SAM was utilized for agricultural parcel boundary extraction in a phenology-aware manner [29]. Furthermore, the same work used fine-tuned Stable Diffusion models to generate realistic synthetic imagery, thus minimizing the need for real image collection [27]. Due to their generic nature, foundation models lack the botanical knowledge to differentiate between crops and weeds; therefore, an interesting hybrid approach was introduced, which combined a lightweight object detector (YOLOv11) with SAM. Specifically, a few-shot fine-tuned YOLOv11 was utilized to generate bounding boxes, which, in turn, guided the segmentation performed by SAM2. This increased the mIOU by +11.3% compared to a baseline while still maintaining a decent speed of 80.02 ms per image [4]. In another hybrid SAM-YOLO approach, a Grounded YOLOv10-SAM2.1 model was used to segment cotton bolls, attaining an mIOU of 86.9% at 18.9 ms per frame (approximately 52 FPS) (pooja et al).

Finally, while the VFMs demonstrated exceptional performance in many vision tasks, they show to have some shortcomings in the agriculture domain. For example, in a setting where the VFM is utilized as a zero-shot anomaly segmenter (i.e. consider weeds as anomalies), such models underperformed severely when facing extreme weed infestations. The reason for the failure was the skewed distribution of anomalies, which constituted the majority of the image in the most infested cases [15]. Lastly, a sanity check was done to see if foundational vision-language models such as Florence2 could be utilized for agriculture tasks without extra finetuning. However, the results showed that such an approach either underperformed significantly or performed worse than random (aashu et al).

3.4 Multimodal Sensor Fusion and the Research Gap

The repeated failures of both conventional CNNs and modern Foundation Models in highly occluded environments are dictated by the same limiting factor: the dependence on standard RGB cameras. Pure optical imagery is inherently limited in its ability to capture morphological details beyond the surface due to varying field illumination, shadows, and similar green hues of different plant species. An RGB-only segmentation model tested on the WeedsGalore dataset scored a dismal mIoU of 63.08% (zeynep et al). Meanwhile, the fusion of different data streams poses its own challenges, including resolution mismatch, extreme temporal displacement between bands, and a lack of adaptive importance weighting (gao et al).

The way to overcome such limitations is to turn to alternative sources of information and attempt Multimodal sensor fusion. By combining the data from standard RGB, Near-Infrared (NIR), Red-Edge (RE), Thermal, or Short-Wave InfraRed (SWIR) sensors, one can obtain a more comprehensive understanding of the target scene. For instance, a simple fusion of RGB, NIR, and RE channels via a Transformer-CNN architecture allowed to boost the segmentation performance to 78.88% mIUO with a mere 8.7 million parameters [18]. Another approach, implemented in the S3-Net framework, demonstrated the effectiveness of combining both RGB texture information and SWIR's ability to peer through vegetation using Vision-Language Models. It achieved an incredible 96.9% accuracy on seed phenotyping at 95 FPS (song et al). Despite the recent progress in self-supervised learning and contrastive vision models (MoCo v2, DenseCL), the majority of the studies show that self-supervised learning underperforms in comparison to conventional supervised training for plant phenotyping tasks. The reason for such a gap is tied to the ability of self-supervised models to capture spatial patterns, which is exacerbated by the high dimensionality and redundancy of plant imagery data [22]. To harness the power of such foundation models for plant phenotyping tasks, one has to implement aggressive model compression techniques. For instance, CDGLYOLOv11Lite utilizes operations fusion and TensorRT's FP16/INT8 precision to achieve 67 FPS on a Jetson Xavier NX while only using 1.9 MB of memory [23].

The analysis of the literature highlights the existence of a major Research Gap: while hybrid Foundation Models such as YOLOSAM address the issue of extensive manual annotation by enabling zero-shot segmentation, and Multimodal fusion helps to alleviate the ambiguity of optical data, the combination of the two has not been explored. In other words, there is little research on how to implement Multimodal fusion in a way that is computationally efficient enough for real-time edge processing. At the same time, most of the existing Foundation Models are optimized for unimodal RGB input. Thus, there is a need to develop a novel efficient multimodal foundation model that would be capable of zero-shot detection of various weed species in different crop fields.

4. OBJECTIVES OF RESEARCH

The objectives of this research are as follows.

1. To develop a software framework that combines a few-shot object detector with a Visual Foundation Model to achieve high-fidelity, zero-shot pixel-level weed segmentation.
2. To systematically test the proposed modal framework across disparate unseen crop

varieties and field environments without explicit retraining.

3. To evaluate and minimize the inference latency and memory footprint of on edge hardware.

5. RESEARCH QUESTION / PROBLEM

Can a hybrid framework integrating a lightweight, few-shot object detector with a visual foundation model support zero-shot weed segmentation across diverse agricultural domains with accuracy comparable to fully supervised models, while reducing computational cost sufficiently for real-time edge deployment? Existing systems rely on domain-specific retraining and exhaustive manual annotations; the absence of a label-efficient, cross-domain foundation model architecture constitutes the research gap addressed here.

6. METHODOLOGY (SAMPLING/MEASUREMENT/RESEARCH DESIGN/DATA ANALYSIS)

The proposed framework comprises a shared convolutional backbone with task-specific necks and three heads: a detection head, an ArcFace-based re-identification head, and a spatio-temporal behaviour head. Tracking is performed downstream using BoT-SORT-style association over the learned embeddings. The overall architecture is shown in Figure-6.1.

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Figure-6.1: Architecture of the proposed unified multi-task framework (Reference if any)

6.1 Datasets

Four public datasets will be used: MultiCamCows2024, Cows2021, CattleBehaviours6, and CVB. Since no single dataset provides all label types per frame, a masked-label training scheme activates only the losses for which ground truth exists in each batch.

6.2 Evaluation Metrics

Detection will be evaluated with mAP; re-identification with rank-1 accuracy and mAP; tracking with HOTA, IDF1, MOTA, and identity switches; behaviour recognition with top-1 accuracy and macro-F1. The plan is summarised in Table-6.1.

Table-6.1: Evaluation metrics and datasets planned for each task of the proposed framework (Reference if any)

Task	Metrics	Dataset
Detection	mAP@0.5, mAP@0.5:0.95	Cows2021
Re-identification	Rank-1, mAP	MultiCamCows2024
Tracking	HOTA, IDF1, MOTA, IDS	Cows2021
Behaviour recognition	Top-1 accuracy, macro-F1	CattleBehaviours6, CVB

6.3 Multi-Task Loss

The framework is trained by minimising a weighted sum of the task losses, where the weights balance gradient magnitudes to mitigate negative transfer:

$$L_{total} = \lambda_{det} L_{det} + \lambda_{id} L_{arc} + \lambda_{beh} L_{beh} \tag{6.1}$$

7. EXPECTED OUTCOMES OF RESEARCH

The research is expected to deliver a jointly optimised, lightweight multi-task framework achieving accuracy comparable to task-specific pipelines at a fraction of the inference cost; empirical evidence on shared versus separate backbones (negative-transfer analysis); a masked-label training recipe for heterogeneous livestock datasets; and a deployable prototype validated on consumer-grade GPU hardware.

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8. APPENDICES

APPENDIX-01

Print of system generated Turnitin similarity report (only ‘originality report’ showing ‘primary sources’) **signed by the author and the supervisor** is required to be placed here.

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